

than three or four lines, so if I were following the guideline “the shorter, the better,” I’d need to split it into even smaller functions!

In this case, the policy of allowing only incredibly short functions might have resulted in simpler functions, but the overall complexity of the program increased drastically. In my opinion, functions should be fewer than 30 lines ideally and definitely no longer than 200 lines. Make your functions as short as reasonably possible but not any shorter.

Function Parameters and Arguments

A function’s parameters are the variable names between the parentheses of the function’s `def` statement, whereas the arguments are the values between a function call’s parentheses. The more parameters a function has, the more configurable and generalized its code can be. But more parameters also mean greater complexity.

A good rule to adhere to is that zero to three parameters is fine, but more than five or six is probably too many. Once functions become overly complicated, it’s best to consider how to split them into smaller functions with fewer parameters.

Default Arguments

One way to reduce the complexity of your function’s parameters is by providing default arguments for your parameters. A *default argument* is a value used as an argument if the function call doesn’t specify one. If the majority of function calls use a particular parameter value, we can make that value a default argument to avoid having to enter it repeatedly in the function call.

We specify a default argument in the `def` statement, following the parameter name and an equal sign. For example, in this `introduction()` function, a parameter named `greeting` has the value `'Hello'` if the function call doesn’t specify it:

```
>>> def introduction(name, greeting='Hello'):
...     print(greeting + ', ' + name)
... 
```